BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

MATHEMATICS

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Mathematics (Tenth, 2022, Lahore Board) (Objective Type) (Group - I)

Time Allowed: 20 Min. | **Maximum Marks:** 15

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. In a ratio x : y, y is called:

(A) Relation
(B) Antecedent
(C) Consequent
(D) Proportion

Q2. Cube roots of − 1 are :

(A) $-1,-\omega,-\omega^2$
(B) $-1,\omega,-\omega^2$
(C) $-1,-\omega,+\omega^2$
(D) $1,-\omega,-\omega^2$

Q3. Roots of the equation $4x^2 - 4x + 1 = 0$ are:

(A) Imaginary
(B) Irrational
(C) Real, equal
(D) Real, unequal

Q4. In continued proportion a : b :: b : c, c is said to be ---- proportional to a and b:

(A) Means
(B) Third
(C) Fourth
(D) Extremes

Q5. The Third Proportional of x^2 and y^2 is:

(A) y^2/x^2
(B) x^2y^2
(C) y^4/x^2
(D) y^2/x^4

Q6. Point (-1, 4) lies in the quadrant:

(A) I
(B) II
(C) III
(D) IV

Q7. The set $\{x|x \in W \wedge x \leq 101\}$ is:

(A) Infinite set
(B) Subset
(C) Null set
(D) Finite set

Q8. If A and B are disjoint sets, then $A \cup B$ is equal to:

(A) A
(B) B
(C) φ
(D) $B \cup A$

Q9. A deviation is defined as a difference of any value of the variable from a:

(A) Constant
(B) Histogram
(C) Sum
(D) Product

Q10. Mean is affected by change in ------:

(A) Place
(B) Value
(C) Rate
(D) Scale

Q11. $1/(1+\sin\theta) + 1/(1-\sin\theta)$ = ------:

(A) $2\sec^2\theta$
(B) $\sec^2\theta$
(C) $2\cos^2\theta$
(D) $\cos\theta$

Q12. The distance of any point of the circle to its centre is called:

(A) Diameter
(B) Radius
(C) A chord
(D) An arc

Q13. Two tangents drawn to a circle from a point outside it are of --- in length:

(A) Half
(B) Equal
(C) Double
(D) Triple

Q14. A 4 cm long chord subtends a central angle of 60°. The radial segment of this circle is:

(A) 1
(B) 2
(C) 3
(D) 4

Q15. The tangent and radius of a circle at the point of contact are ------ :

(A) Parallel
(B) Not perpendicular
(C) Perpendicular
(D) None

Mathematics (Tenth, 2022, Lahore Board) (Essay Type) (Group - I)

Time Allowed: 2.10 hrs. | **Maximum Marks:** 60

QPART - I.

QQ.2. (Marks 12) Write short answers to any SIX (6) questions:

(i) Write the quadratic equation in standard form: $(x + 7)(x - 3) = -7$

(ii) Solve the equation by quadratic formula: $2 - x^2 = 7x$

(iii) Define quadratic equation.

(iv) Write the quadratic equation from given roots : $0, -3$

(v) Evaluate : $(1 - \omega - \omega^2)^7$

(vi) Find ω^2 if $\omega = (-1 + \sqrt{-3})/2$

(vii) If $3(4x - 5y) = 2x - 7y$, find the ratio $x : y$

(viii) Find a mean proportional between 20, 45

(ix) Define ratio.

QQ.3. (Marks 12) Write short answers to any SIX (6) questions:

(i) Define proper fraction and give one example.

(ii) Resolve into partial fractions: $1/(x^2 - 1)$

(iii) If $X = \{1, 4, 7, 9\}$ and $Y = \{2, 4, 5, 9\}$, then find $Y \cup X$.

(iv) If $A = \{0,2,4\}$, $B = \{-1, 3\}$ then find $A \times B$

(v) Define an One-one function.

(vi) Write all the subsets of {a, b}

(vii) Define geometric mean.

(viii) Find arithmetic mean for the set of data: 12, 14, 17, 20, 24, 29, 35, 45

(ix) Find the range of the data: 11500, 12400, 15000, 14500, 14800

QQ.4. (Marks 12) Write short answers to any SIX (6) questions:

(i) Convert $3\pi/4$ into degrees.

(ii) Find θ, when $\ell = 2$ and $r = 3.5$ cm.

(iii) Prove that $(1 - \sin^2\theta)(1 + \tan^2\theta) = 1$

(iv) How many minutes are there in two right angles?

(v) Define acute angle.

(vi) Define secant.

(vii) Define circumference of a circle.

(viii) Divide an arc of any length into two equal parts.

(ix) Define inscribed circle.

Mathematics (Tenth, 2022, Lahore Board) (PART-II) (Group - I)

Time Allowed: 2.10 hrs. | **Maximum Marks:** 60

Note: Attempt THREE questions in all. But question No. 9 is Compulsory.

QQ.5(a). (Marks 4) Solve the equation: $\sqrt{(3x + 100)} - x = 4$

QQ.5(b). (Marks 4) If α, β are the roots of the equation $x^2 + px + q = 0$, then evaluate $\alpha^2 + \beta^2$.

QQ.6(a). (Marks 4) Using the theorem of componendo-dividendo, find the value of $(m+5n)/(m-5n) + (m+5p)/(m-5p)$ if $m = 10np/(n+p)$

QQ.6(b). (Marks 4) Resolve into partial fractions: $1/((x - 1)^2(x + 1))$

QQ.7(a). (Marks 4) If U = {1,2,3,4,5,6,7,8,9,10}, A = {1,3,5,7,9} and B = {2,3,5,7} then prove that $(A \cup B)' = A' \cap B'$

QQ.7(b). (Marks 4) Find the mean length and standard deviation: [Length | 20-22 | 23-25 | 26-28 | 29-31 | 32-34; Frequency | 3 | 6 | 12 | 9 | 2]

QQ.8(a). (Marks 4) Prove that : $(1+\sin\theta)/(1-\sin\theta) - (1-\sin\theta)/(1+\sin\theta) = 4\tan\theta \sec\theta$

QQ.8(b). (Marks 4) Circumscribe a circle about a triangle ABC with sides: |AB| = 6 cm, |BC| = 3 cm, |CA| = 4 cm

QQ.9. Prove that perpendicular from the centre of a circle on a chord bisects it. OR Prove that the measure of a central angle of a minor arc of a circle is double that of the angle subtended by the corresponding major arc.

GROUP – II

Mathematics (Tenth, 2022, Lahore Board) (Objective Type) (Group - II)

Time Allowed: 20 Min. | **Maximum Marks:** 15

> Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. An equation of the type $3^x + 3^{2-x} + 6 = 0$ is a/an:

(A) Exponential equation
(B) Radical equation
(C) Reciprocal equation
(D) Quadratic equation

Q2. Two square roots of a unity are:

(A) 1, -1
(B) 1, ω
(C) 1, −ω
(D) ω, ω^2

Q3. If α, β are the roots of $px^2 + qx + r = 0$, then sum of the roots 2α and 2β is:

(A) −q/p
(B) r/p
(C) −2q/p
(D) q/2p

Q4. If u/v = v/w = k , then :

(A) $u = wk^2$
(B) $u = vk^2$
(C) $u = w^2k$
(D) $u = v^2k$

Q5. In a ratio x : y, y is called:

(A) Relation
(B) Antecedent
(C) Consequent
(D) Proportion

Q6. $(x^3 + 1)/((x - 1)(x + 2))$ is:

(A) A proper fraction
(B) An improper fraction
(C) An identity
(D) A constant term

Q7. The number of elements in power set of {1, 2, 3} is:

(A) 4
(B) 6
(C) 8
(D) 9

Q8. Point (-1, 4) lies in the quadrant:

(A) I
(B) II
(C) III
(D) IV

Q9. A frequency polygon is a many sided:

(A) Closed figure
(B) Rectangle
(C) Square
(D) Triangle

Q10. Mean is affected by change in -----:

(A) Place
(B) Value
(C) Rate
(D) Scale

Q11. 1/2 cosec 45° = ------:

(A) 1/(2√2)
(B) 1/√2
(C) √2
(D) √3/2

Q12. A chord passing through the centre of a circle is called:

(A) Secant
(B) Diameter
(C) Circumference
(D) Radius

Q13. Two tangents drawn to a circle from a point outside it are of ----- in length:

(A) Half
(B) Equal
(C) Double
(D) Triple

Q14. A pair of chords of a circle subtending two congruent central angles is:

(A) Parallel
(B) Incongruent
(C) Overlapping
(D) Congruent

Q15. How many tangents can be drawn from a point outside the circle:

(A) 4
(B) 3
(C) 1
(D) 2

Mathematics (Tenth, 2022, Lahore Board) (Essay Type) (Group - II)

Time Allowed: 2.10 hrs. | **Maximum Marks:** 60

QPART - I.

QQ.2. (Marks 12) Write short answers to any SIX (6) questions:

(i) Write the quadratic equation in standard form: $(x+4)/(x-2) - (x-2)/x + 4 = 0$

(ii) Solve the equation by quadratic formula: $4x^2 - 14 = 3x$

(iii) Define radical equation.

(iv) Write the quadratic equation from given roots : $3 + \sqrt{2}, 3 - \sqrt{2}$

(v) Evaluate : $(2 + 2\omega - 2\omega^2)(3 - 3\omega + 3\omega^2)$

(vi) Find the sum and product of the roots of equation $2px^2 + 3qx - 4r = 0$

(vii) If $y \propto x$ and $y = 7$, when $x = 3$, find y in terms of x.

(viii) Find a third proportional to $a^2 - b^2$, $a - b$

(ix) Define inverse variation.

QQ.3. (Marks 12) Write short answers to any SIX (6) questions:

(i) What is proper fraction?

(ii) Resolve into partial fractions: $(7x - 9)/((x + 1)(x - 3))$

(iii) If X = {1, 4, 7, 9} and Y = {2, 4, 5, 9}, then find X ∪ Y.

(iv) Find a, b if (a − 4, b − 2) = (2,1)

(v) If L = {a, b, c} then find L × L.

(vi) Define subset and give an example.

(vii) Define geometric mean.

(viii) Find the range of the data: 11500, 12400, 15000, 14500, 14800

(ix) What is histogram?

QQ.4. (Marks 12) Write short answers to any SIX (6) questions:

(i) Define radian.

(ii) Find r, when $\ell = 52$ cm, $\theta = 45°$.

(iii) Prove that: $(\sin\theta + \cos\theta)/\cos\theta = 1 + \tan\theta$

(iv) Write the relation between degree and radian.

(v) Define tangent of a circle.

(vi) Define circum circle.

(vii) Define chord of a circle.

(viii) Define central angle.

(ix) The length of each side of a regular octagon is 3 cm. Measure its perimeter.

Mathematics (Tenth, 2022, Lahore Board) (PART-II) (Group - II)

Time Allowed: 2.10 hrs. | **Maximum Marks:** 60

Note: Attempt THREE questions in all. But question No. 9 is Compulsory.

QQ.5(a). (Marks 4) Solve the equation by completing square: $ax^2 + 4x - a = 0$ $a \neq 0$

QQ.5(b). (Marks 4) Show that the roots of the equation $(b - c)x^2 + (c - a)x + (a - b) = 0$ are real.

QQ.6(a). Find x in the proportion $8 - x : 11 - x :: 16 - x : 25 - x$ (4)

QQ.6(b). (Marks 4) Resolve into partial fractions: $(x - 5)/(x^2 + 2x - 3)$

QQ.7(a). (Marks 4) If U = {1,2,3,4,5,6,7,8,9,10}, A = {1,3,5,7,9} and B = {2, 3, 5, 7} then prove that $(A \cap B)' = A' \cup B'$

QQ.7(b). (Marks 4) Find the standard deviation "S": 9, 3, 8, 8, 9, 8, 9, 18

QQ.8(a). (Marks 4) If $\tan\theta = 4/3$ and $\sin\theta < 0$, then find values of other trigonometric functions at θ.

QQ.8(b). Circumscribe a circle with regard to a right angle triangle with sides 3 cm, 4 cm and 5 cm.

QQ.9. Prove that a straight line, drawn from the centre of a circle to bisect a chord (which is not a diameter) is perpendicular to the chord. OR Prove that the measure of a central angle of a minor arc of a circle is double that of the angle subtended by the corresponding major arc.